

Collective Properties of Biological Systems

Solitons and Coherent Electric Waves in a Quantum Field Theoretical Approach

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ABSTRACT: We present a dynamical scheme for biological systems. We use methods and techniques of quantum field theory since our analysis is at a microscopic molecular level. Davydov solitons on biomolecular chains and coherent electric dipole waves are described as collective dynamical modes. Electric polarization waves predicted by Fröhlich are identified with the Goldstone massless modes of the theory with spontaneous breakdown of the dipole-rotational symmetry. Self-organization, dissipativity, and stability of biological systems appear as observable manifestations of the microscopic quantum dynamics.

1. Introduction

Biological systems are collective systems indeed. They seem to be satisfactorily described when the list of their components is supplemented by the prescription of their location inside the system and the timing of their operation. Biologists refer to such a timing as a “cell clock”; what is now

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required is the engine responsible for such a dynamical ordering. Dynamical mechanisms that could give account for collective behavior have been proposed in the frame of different approaches.

H. Fröhlich⁽¹⁾ has introduced, in a framework of far-from-equilibrium processes, coherent electric polarization waves as the physical agent able to control the working of distant and separate parts of the system, making them cooperative. On the other hand, it has been shown that biomolecules are able to host on their own chains, in a conservative framework, highly nonlinear subdynamics which give rise to deep structural and conformational changes.

Very interesting subdynamics have been investigated by A. S. Davydov,^(2,3) who has shown that metabolic reactions can result in strictly confined "bumps" traveling along biomolecular chains and carrying energy in an efficient way a long distance away from the point of release.

Our aim is to provide a systematic approach to the dynamics of a biological system^(4,5) by using the widely accepted tool for exploring the collective properties of a system made from a large number of elementary microscopic components, namely, quantum field theory (QFT). We choose quantum field theory instead of classical field theory since quantum theory has proved to be the most adequate tool for describing atoms, molecules, and their interactions, and we wish actually to carry out an investigation of the biological system at a molecular level. Moreover, we choose quantum field theory instead of quantum mechanics because we are not interested in singling out just a few degrees of freedom of the system; on the contrary we would like to get a comprehensive description of the dynamics of the whole system (collective behavior).

The QFT approach could be sketched as follows. In a preliminary stage the microscopic components of the system are enumerated; the state of minimum energy is called the ground state or the vacuum. It is possible that more than one ground state exists for the same system, each of them being characterized by different macroscopic properties. In such a case different regimes or phases will correspond to different vacua. Subsequently, the microscopic components are linked to field operators whose dynamical evolution is obtained through equations of motion derived from a Lagrangian. The usual prescription is to choose this Lagrangian as symmetric as possible in order to get the simplest equations of motion while that part of the dynamics which is less symmetric is taken into account via the vacuum.

Where the vacuum does not exhibit the same invariance of the Lagrangian, one says that a spontaneous breakdown of the symmetry occurs. General theorems have been derived which elucidate the dynamical consequences on the whole system of the interplay between a symmetric Lagrangian and a nonsymmetric vacuum. The Goldstone theorem⁽⁶⁾ is a

very important one; it states that massless modes (usually called Goldstone bosons) appear when the Lagrangian is invariant, while the vacuum is non-invariant under a certain group of transformations. Notice that the invariance or noninvariance of the Lagrangian and of the vacuum under a transformation compels us to take into account the whole of the system degrees of freedom so that the Goldstone modes are really collective modes. A massless mode actually describes a long-range interaction among the system components and such an interaction establishes a correlation (i.e., order) among the components. The noninvariance of the vacuum under a certain transformation thus implies that this vacuum is an ordered state, in which absorption and emission of quanta of the Goldstone field, undergoing the transformation of noninvariance, take place. The mathematical structure of the theorem makes evident that the above-mentioned boson quanta exist in a condensed form and their condensation onto the ground state corresponds to a sort of "printing" of the symmetry pattern on the vacuum.

It is interesting to compare the collective properties arising from a microscopic theory with the macroscopic predictions obtained at a thermodynamic level in the framework of different approaches. The ordering of biological systems by coherent electric waves as proposed by Fröhlich could be understood, for instance, as the phenomenological manifestation of the breakdown of some symmetry at a microscopic level.^(4,5,7) In this case the fact that the condensation, i.e. the emission and absorption, of Goldstone boson quanta does not require any energy expense is the microscopic counterpart of the thermodynamic dissipativity condition, which is a necessary condition⁽¹⁾ for the appearance of these waves. Dissipativity is a very general and important feature of living systems as far-from-equilibrium systems.⁽⁸⁾ If we consider the Goldstone modes not as particles but as waves, the condensation of quanta producing an ordered state would lead us to take into account coherent waves. Coherent electric polarization waves require then, in the above framework, a vacuum with a nonzero value for the relevant field, namely, for the polarization field. As a matter of fact, water around biomolecules exhibits a nonzero polarization, i.e., an electret.^{(9,10)*} Zero frequency modes have been also observed, which point to the existence of massless modes.⁽¹¹⁾

Nonlinear dynamical effects on one-dimensional systems such as biomolecular chains, e.g., Davydov solitons, can also be analyzed in terms of boson condensation.⁽¹²⁻¹⁴⁾ In the Fröhlich wave case, the condensation

* We define the electret state of a dielectric material, in agreement with the definition of Refs. 9 and 10, as a metastable state characterized by a stored polarization. Its decay time must be assumed long with respect to the characteristic time of experiments performed on the material.

induced by a symmetry breakdown occurs on large regions homogeneously, whereas in one-dimensional molecular systems one has quite localized condensations producing solitons. QFT can provide a unified description of such physically different situations. It is also interesting to note that an analysis by means of boson condensation can be performed in a variety of systems in many-body physics.⁽¹²⁾ Thus a QFT approach has the appealing feature of extending methods and mathematics currently used in condensed matter physics to living matter.

In the following sections we present such a unified treatment. First we sketch qualitatively the interplay between Davydov and Fröhlich mechanisms (Section 2). Then we present an account of both of them and we discuss the transition from one regime to the other (Section 3). Finally we discuss the phenomenological implications of the theoretical scheme and the available supporting experimental evidence (Section 4).

2. Solitons, Electrets, and Fröhlich Waves

A very appealing schematization features a biological system as a set of electric dipoles, a dielectric with peculiar properties. Both water molecules—which account for a large amount of the global weight—and the macromolecules having a role in the biochemical machinery (DNA, RNA, proteins, and so on) have relevant dipole values or exhibit atomic subgroups with large dipole values. The starting point of the Fröhlich approach, for instance, is just to treat living matter as a dielectric, where the correlations among different parts are established by electric polarization waves. On the other hand the dynamics of a one-dimensional set of dipoles gives rise to vibrational solitons on biomolecular chains as proposed by Davydov.

Let us first discuss these two mechanisms separately. Biological systems are considered by Fröhlich as three-dimensional sets of dipoles. It is well known that the basic dipole–dipole interaction is rotationally invariant, so that we expect that the Lagrangian describing the whole system—whatever its detailed mathematical form is—be rotationally invariant too. On the other hand an electret (not a ferroelectret) has been detected for almost all the biomolecules.^(9,10) Furthermore, this electret does not belong to the bare molecular structure but to its “dressing,” the surrounding hydration water. Some sort of molecular dynamics produces a nonzero electric polarization in the surroundings, thus breaking the rotational symmetry of the dipole dynamics. It is possible consequently to introduce the formalism and theorems concerning spontaneously broken symmetries and their dynamical rearrangement (see Section 3).⁽¹⁵⁾ In a first rough treatment one could use the Goldstone theorem, which predicts the

appearance of massless bosons. A more adequate treatment would consider the electric field as a gauge field; a massive electric quantum excitation would then appear. It is known that the mass of a quantum of a field is inversely proportional to its propagation range; consequently, while a rigorously massless mode would correspond to an infinite propagation range (infinite size of the system), a nonzero mass corresponds to the observed finite range correlations (finite size of the system). A further improvement would concern the introduction of finite temperature effects which will allow thermal excitations. The problem will then arise of the interaction of coherent electric waves with thermal effects.

Let us now consider the one-dimensional dynamics occurring on the macromolecular chains. Davydov has shown^(2,3,16) that a vibrational soliton—namely, a localized excitation strictly bound to a localized deformation—occurs on one-dimensional chains made by weakly bound monomers under two conditions:

- i. the initial excitation is localized, i.e., energy is supplied to the system at small sized sites;
- ii. the coupling between monomers exhibits a positive anharmonicity.

Under the above conditions solitons are created and travel quite slowly on the chain exhibiting a remarkable stability when their lifetime is compared with the one of the original excitation. Solitons have been therefore proposed as the long-range carriers of the energy produced by metabolic reactions and stored on the molecules.⁽¹⁷⁾ For this reason they were used in a model of muscle contraction by Davydov.

Apart from the storage of energy, many additional and partially unexplored consequences arise on the biomolecules because of solitons. Conformational and structural changes may occur as well as changes in the vibrational frequency patterns. The binding of hydration water molecules to biomolecules is also affected. Davydov has recently pointed out⁽¹⁸⁾ that a soliton on the α -helix of a protein behaves as a mobile trap for loosely bound electrons, giving rise to an electrosoliton which has the features of a supercurrent. In other words, once a vibrational soliton is produced when a certain amount of energy is supplied, an electric current is obtained without further expense of energy. It is reasonable to assume that such a supercurrent, via its associated fields, might induce a preferred direction (an orientation) among water molecules surrounding the biomolecule. Actually, it is a fact⁽⁹⁾ that the activation energy that one must supply to a biomolecule in order that an electret may appear around it is just of the same order of magnitude (0.2–0.4 eV) as the energy required for the production of a soliton on an α -helix protein through the ATP hydrolysis.

The above-discussed mechanisms are particular ones. It is possible to integrate them in a provisional, unified scheme giving a partial qualitative

picture of cell operation.⁽¹⁹⁾ The metabolic reaction, governed by nutrition and membrane activity, produces localized excitations at selected points of selected molecules (DNA). Such excitations propagate in soliton form along the chains storing energy on them; the rate of accumulation is controlled by the difference between the number of solitons produced per unit time (metabolic reaction rate) and the number of solitons decaying in unit time. This rate is linked to the chain elasticity, a time-dependent parameter. We have described what we call the energy charge phase of the cell cycle. Simultaneously, a water electret is created around the molecule because of the soliton accumulation. When the electret reaches a threshold corresponding to the breakdown of the rotational symmetry of the dipole set (representing a finite-sized cell), then a different dynamical regime is switched on, where electric polarization waves must appear according to the previous discussion on the dynamical rearrangement of symmetry. These waves, propagating in the system, are instrumental in its biochemical organization, as discussed elsewhere.^(19,20)

3. *Boson Condensation and Collective Modes*

Quantum field theory provides a two-level description for a system:

- a. the dynamical level, where basic microscopic fields, say $\psi(x)$, are introduced, which are called the Heisenberg fields and obey the dynamic equations

$$A(\partial) \psi(x) = J[\psi(x)] \quad (3.1)$$

- b. the phenomenological level, where observable quantities are expressed in terms of physical fields, say $\phi(x)$, which are called also asymptotic or quasiparticle fields and satisfy free field equations

$$K(\partial) \phi(x) = 0 \quad (3.2)$$

In Eqs. (3.1) and (3.2), $A(\partial)$ and $K(\partial)$ are kinetic differential operators, x stands for $(\mathbf{x}, t)$, and $J[\psi(x)]$ is a nonlinear functional of $\psi(x)$.

As a next step, the set of the physical states, or Fock space, of the system is introduced. It is denoted by $\{|0\rangle\}$ and contains all the states obtained by repeated application of the “quasiparticle” creation operators on the ground state, or “vacuum,” $|0\rangle$. The “quasiparticle” creation operators are functions of the fields $\phi(x)$.

At the first level of description the “forces” acting among the system components appear through the functional $J[\psi(x)]$, while, at the second level, the fields are considered free, because only fairly independent quan-

tities are observable. In a scattering process, for instance, the particles are observed only before and after the collision. In a complex system either the whole compound system or the correlations among the constituents—namely, the free particles acting as messengers (phonons and so on)—are observed. These “free” phenomenological fields are of course the output of the inner dynamics described at the first level and such relationship is provided by the “dynamical map”^(12,15):

$$\langle a | \psi(x) | b \rangle = \langle a | \Psi[\phi(x)] | b \rangle \quad (3.3)$$

Equation (3.3) is the bridge between the deep underworld where the “forces” are at work and the apparent world where observations are performed. In Eq. (3.3) $|a\rangle$ and $|b\rangle$ are wave-packet states, which belong to the Fock space $\{|0\rangle\}$ and $\Psi[\phi(x)]$ is a functional of (normally ordered) products of the $\phi(x)$ fields.

Let us now investigate how a symmetry at the dynamical level would affect the phenomenological description. Consider a transformation which leaves Eq. (3.1) invariant, namely, let g be an element of the symmetry group of Eq. (3.1):

$$\psi'(x) = g\psi(x) \quad (3.4)$$

$$A(\partial) \psi'(x) = J[\psi'(x)] \quad (3.5)$$

The dynamical map (3.3) would then select a field $\phi'(x)$ corresponding to $\psi'(x)$, defining a transformation h which would give $\phi'(x)$ as a function of the field $\phi(x)$ corresponding to the original $\psi(x)$ via the dynamical map

$$\langle a | g\psi(x) | b \rangle = \langle a | \Psi[h\phi(x)] | b \rangle \quad (3.6)$$

Usually the functional $\Psi[\phi(x)]$ is highly nonlinear, so that h is expected to be different from g . However, since g leaves Eq. (3.1) invariant (i.e., it corresponds to a physical process which requires no expense of energy), h leaves in turn Eq. (3.2) invariant

$$\phi'(x) = h\phi(x) \quad (3.7)$$

$$K(\partial) \phi'(x) = 0 \quad (3.8)$$

If the transformations g form a group G , then the transformations h form a group H , too.

We are interested in the situation $H \neq G$, namely, when the dynamical map (3.6) rearranges the symmetry transformations so that the system acquires a symmetry at the phenomenological level, which is quite different from the one at the microscopic dynamical level. In this case we say that a dynamical rearrangement of the symmetry occurs.

We are going to explore the consequences of the dynamical rearrangement of the symmetry in the two cases of the Davydov vibrational solitons on macromolecules and of the Fröhlich electric waves in the whole biological system. In the first case we deal with an explicit form of the equations, so that a complete treatment is possible, while in the second case we limit ourselves to the consideration of the dynamical rearrangement of the symmetry and of the consequent appearance of massless modes.

Let us consider first the soliton dynamics on the biomolecular chains.

3.1. Davydov Soliton Regime

A biomolecular chain can be schematized as a one-dimensional chain of weakly bound monomers of mass M , whose electric dipole moment is aligned along the chain. Consequently the internal dipole quantum number, giving the spatial orientation of the dipole, can be taken as frozen; the electric dipole, which would be described by a spinor in the usual three dimensions, is described in this one-dimensional case by a scalar.

In a first approximation the molecule length is assumed infinite and corrections due to the finite length are introduced as a second step. The intramolecular excitation brought by an external supply of energy (e.g., the output of a chemical reaction) is described by a complex Heisenberg boson field $\psi(\xi, t)$, solution of a nonlinear Schrödinger equation^(2,3,16):

$$\left(i\hbar \frac{\partial}{\partial t} - A + J \frac{\partial^2}{\partial \xi^2} \right) \psi(\xi, t) = -G\psi^+(\xi, t)\psi(\xi, t)\psi(\xi, t) \quad (3.9)$$

The right-hand side of Eq. (3.9) is understood to be normally ordered. A is a constant energy term which may depend on the free exciton energy, on the static interaction energy, and on the deformation energy. J is the resonance interaction energy between neighboring monomers. The quasiparticle free equation (3.2) becomes in this case

$$\left(i\hbar \frac{\partial}{\partial t} - A + J \frac{\partial^2}{\partial \xi^2} \right) \phi(\xi, t) = 0 \quad (3.10)$$

The operator $\phi(\xi, t)$ is the free exciton field. As a first step we are going to show that the c -number Davydov soliton^(2,3,16)

$$\chi_c(\xi, t) = \left(\frac{\mu}{2} \right)^{1/2} \frac{\exp i \left[\frac{\hbar v}{2J} (\xi - \xi_0) - E_v \frac{t}{\hbar} \right]}{\cosh \mu(\xi - \xi_0 - vt)} \quad (3.11)$$

$$\int |\chi_c(\xi, t)|^2 d\xi = 1 \quad (3.12)$$

is just a convenient vacuum expectation value for $\psi(\xi, t)$ in the “classical” limit. In Eq. (3.11) $\hbar$ is the Planck constant and

$$E_v = A + \frac{\hbar v^2}{4J} - J\mu^2, \quad \mu = \frac{G}{4J} \quad (3.13)$$

Let us assume a transformation B such that

$$\psi(\xi, t) \rightarrow \psi'(\xi, t) = B\psi(\xi, t) \quad (3.14)$$

$$\chi_c(\xi, t) = \langle 0 | \psi'(\xi, t) | 0 \rangle |_{\hbar=0} = \langle 0 | B\psi(\xi, t) | 0 \rangle |_{\hbar=0} \quad (3.15)$$

which leaves Eq. (3.9) invariant. The dynamical map (3.3) then becomes

$$\langle 0 | B\psi(\xi, t) | 0 \rangle = \langle 0 | \psi[\beta\phi(\xi, t)] | 0 \rangle \quad (3.16)$$

which defines β as an invariant transformation for Eq. (3.10). Let us require also that β be a canonical transformation, namely, β preserves the canonical commutation relations

$$\begin{aligned} [\phi(\xi, t), \dot{\phi}(\xi', t)] &= [\phi'(\xi, t), \dot{\phi}'(\xi', t)] = i\delta(\xi - \xi') \\ [\phi(\xi, t), \phi(\xi', t)] &= [\phi'(\xi, t), \phi'(\xi', t)] = 0 \\ [\phi(\xi, t), \phi(\xi', t)] &= [\phi'(\xi, t), \phi'(\xi', t)] = 0 \end{aligned} \quad (3.17)$$

where

$$\phi'(\xi, t) = \beta\phi(\xi, t) \quad (3.18)$$

We consider now the general class of canonical transformations

$$\phi'(\xi, t) = \beta\phi(\xi, t) = \phi(\xi, t) + f(\xi, t) \quad (3.19)$$

where the field $\phi(\xi, t)$ is translated by a c -number function $f(\xi, t)$. The transformation (3.19) would leave Eq. (3.10) invariant, provided that $f(\xi, t)$ is a (c -number) solution of the same equation, namely,

$$f(\xi, t) = ce^{-\mu(\xi - vt)} \exp \left[i \left(\frac{\hbar v}{2J} \xi - \frac{E_v t}{\hbar} \right) \right] \quad (3.20)$$

Let us show that the transformation (3.19) on the fields ϕ , which corresponds to the “translation” (3.20), induces through the dynamical map just the transformation (3.14) on the fields ψ , so that Eq. (3.15) is satisfied. The generator of (3.19) in the $v=0$ frame of reference is

$$D = - \int_{-\infty}^{\infty} d\xi [g(\xi, t) \dot{\phi}(\xi, t) - \dot{g}(\xi, t) \phi(\xi, t)] \quad (3.21)$$

where

$$\begin{aligned} g(\xi, t) &= \theta(\xi) f_0(\xi, t) \\ f_0(\xi, t) &= f(\xi, t) |_{v=0} \end{aligned} \quad (3.22)$$

$\theta(\xi)$ is the usual step function: $\theta(\xi) = 0$ for $\xi < 0$ and $\theta(\xi) = 1$ for $\xi > 0$. It is possible to read Eq. (3.19) as

$$\phi'(\xi, t) = \beta \phi(\xi, t) = \lim_{g \rightarrow f_0} e^{-iD} \phi(\xi, t) e^{iD} \quad \text{at } v=0 \quad (3.23)$$

and

$$\langle 0 | e^{-iD} \phi(\xi, t) e^{iD} | 0 \rangle |_{g=f_0} = \langle f_0 | \phi(\xi, t) | f_0 \rangle = f_0(\xi, t) \quad (3.24)$$

The state $|f_0\rangle$ is defined as

$$|f_0\rangle \equiv e^{iD} |0\rangle |_{g=f_0}, \quad \langle f_0 | f_0 \rangle = 1 \quad (3.25)$$

We introduce now the annihilation operator $a_{\bar{k}}$ of a quantum of the field $\phi(\xi, t)$ whose momentum is $\bar{k}$; let us also consider the c -number function $\alpha(\bar{k})$, the Fourier transform of a function which coincides with $f_0(\xi, t)$ in a suitable ξ domain; then

$$a_{\bar{k}} |f_0\rangle = \alpha(\bar{k}) |f_0\rangle \quad (3.26)$$

i.e., $|f_0\rangle$ is a coherent state for $\phi(\xi, t)$, namely, any functional of ϕ acts on $|f_0\rangle$ as a c -number.

The dynamical map (3.3) then becomes

$$\langle f_0 | \psi(\xi, t) | f_0 \rangle = \Psi[f_0(\xi, t)] \quad (3.27)$$

$$\langle f_0 | J[\psi(\xi, t)] | f_0 \rangle = J[\Psi[f_0(\xi, t)]] \quad (3.28)$$

A boost to the original frame of reference and the substitution of Eqs. (3.27) and (3.28) into Eq. (3.9) produce at last

$$\left(i\hbar \frac{\partial}{\partial t} - A + J \frac{\partial^2}{\partial \xi^2} \right) \psi[f] = -G |\psi[f]|^2 \psi[f] \quad (3.29)$$

Equation (3.29) is a c -number equation which coincides with the one proposed by Davydov [see Eq. (2.10) of Ref. 2]. The solution of (3.29) is, then, just the soliton (3.11)

$$\Psi[f(\xi, t)] = \chi_c(\xi, t) \quad (3.30)$$

which allows us to determine the constant c in Eq. (3.20):

$$c = (2\mu)^{1/2} e^{\mu\xi_0} \exp\left(-i \frac{\hbar v}{2J} \xi_0\right) \quad (3.31)$$

In conclusion from the above formalism we can deduce that the Davydov soliton is the macroscopic envelope of the localized boson condensation of the excitation quanta $\phi(\xi, t)$ induced by the β transformation (3.19).⁽¹²⁻¹⁴⁾ The vanishing of the coefficient G of the nonlinear term of Eq. (3.29), via Eq. (3.13), implies the disappearance of the soliton, which then looks like a necessary consequence of the nonlinearities of the dynamics.

As a further step, we are going now to show how the QFT approach could recover all the relevant Davydov results starting from a restricted set of assumptions.⁽⁴⁾

We started actually from the nonlinear Schrödinger equation (3.9) governing the $\psi(\xi, t)$ field only. A deformation (phonon) field has not, so far, been independently introduced. But the density probability ρ^f of ψ in the soliton state $|f\rangle$

$$\rho^f(\xi, t) = \gamma \langle f | \psi^+(\xi, t) \psi(\xi, t) | f \rangle = \gamma |\chi_c(\xi, t)|^2 \quad (3.32)$$

or [see Eq. (3.11)]

$$\rho^f(\xi, t) = \gamma \frac{\mu}{2} \frac{1}{\cosh^2 \mu(\xi - \xi_0 - vt)} \quad (3.33)$$

is a solution of the (phonon) equation

$$\left(\frac{\partial^2}{\partial t^2} - v^2 \frac{\partial^2}{\partial \xi^2} \right) \rho^f(\xi, t) = 0 \quad (3.34)$$

The phonon equation is then a direct consequence of the self-interacting intramolecular excitation field ψ , which is the only independent physical object. As a matter of fact, the ψ -excitation field dynamics induces conformational changes in the chain lattice which are described by a phonon field.

It is useful to recast Eq. (3.34) in the more familiar form of the Davydov Eq. (2.56) of Ref. 2. Let us define

$$\begin{aligned} v_{ac} &= \left(\frac{W}{M} \right)^{1/2} \\ v^2 &= v_{ac}^2 - v_{ac}^2 (1 - s^2) \\ s &= \frac{v}{v_{ac}} \ll 1 \end{aligned} \quad (3.35)$$

where w is the chain elasticity coefficient and v_{ac} is the sound velocity on the chain in absence of solitons. Equation (3.34) is consequently transformed into

$$\left(\frac{\partial^2}{\partial t^2} - v_{ac}^2 \frac{\partial^2}{\partial \xi^2} \right) \beta^f(\xi, t) - \frac{2K}{M} \frac{\partial}{\partial \xi} |\chi_c(\xi, t)|^2 = 0 \quad (3.36)$$

where

$$K = 1/2w(1 - s^2) \gamma \quad (3.37)$$

$$-\frac{\partial}{\partial \xi} \beta^f(\xi, t) = \rho^f(\xi, t) \quad (3.38)$$

Equation (3.36) is just the Davydov equation (2.56) of Ref. 2. Moreover Eq. (3.29) can be transformed as follows:

$$\left[i\hbar \frac{\partial}{\partial t} - A + J \frac{\partial^2}{\partial \xi^2} - 2K \frac{\partial \beta^f(\xi, t)}{\partial \xi} \right] \chi_c(\xi, t) = 0 \quad (3.39)$$

where

$$\mu = \frac{K^2}{w(1 - s^2) J} \quad (3.40)$$

Equation (3.39) coincides with Eq. (2.5a) of Ref. 2. We conclude that the present QFT approach is equivalent to the Davydov analysis. All the results of the latter theory can be recovered in the QFT framework, such as, for instance, the soliton stability, the soliton-photon interaction and the attraction of electrons by the self-generating potential wells leading to quasisuperconducting properties of the molecular chains, hosting solitons. The chains considered above were assumed to be infinitely long; let us now drop this assumption. This change affects mainly the stability of the soliton. The soliton is stable when the local deformation $\beta^f(\xi, t)$ [or $\rho^f(\xi, t)$] is stable and vice versa [see Eqs. (3.36) and (3.39)]. The conditions for the stability of β^f have been investigated in Ref. 14. For the sake of conciseness, we do not report here such an analysis (see also Ref. 4), but we give the conclusion. In a finite chain the soliton state has a nonzero transition amplitude with any state of n free excitation quanta, so that its lifetime has to be finite. The longer the chain, the longer the lifetime.

Very important features of solitons, such as their rate of creation and decay, their density on the chain, the effects of external perturbations (for instance, changes of the soliton density and decay rate induced by variations of the physical and chemical structure of the chain) are yet a matter of investigation.

All the above analysis shows that an incoherent energy supply can be transformed into a coherent but localized form (the soliton) through the

inner nonlinear dynamics occurring on the molecules constituting the biological system. The next step is the dynamical appearance of self-organization in the system without further expense of energy; energy stored in soliton form during the “charge” or Davydov regime is completely released outward during the “discharge” or Fröhlich regime.

3.2. Fröhlich Regime

It has been experimentally found that water surrounding biomolecules can be brought in an electret state by low electric fields at low frequency.

We have said in Section 2 that the electret activation energy is of the same order of magnitude as the energy required for the production of a soliton on a biomolecule. The numerical coincidence suggests that the soliton might induce the observed electret while traveling on the chain. How it could happen is at present under investigation; we limit ourselves here to suggesting that the soliton could be a trigger for the water polarization. As a consequence the dipole quantum number carried by the dipole excitation ψ field would no longer be frozen since the water electret has transformed the previously one-dimensional system into a three-dimensional one. Let us describe the dipole quantum number by the spinor:

$$\psi(x) = \begin{pmatrix} \psi\uparrow(x) \\ \psi\downarrow(x) \end{pmatrix} \quad (3.41)$$

where $x \equiv (\tilde{x}, t)$. The Lagrangian $\mathcal{L}$ of the whole biological system is very complicated, but it is safe to assume its invariance under the rotation of the dipole degree of freedom of the doublet field (3.4). More technically, we assume its invariance under the $SU(2)$ group of transformations

$$\psi(x) \rightarrow \psi'(x) = e^{i\theta_i \sigma_i} \psi(x), \quad i = 1, 2, 3 \quad (3.42)$$

θ_i are real parameters and σ_i are the Pauli matrices. Let us now get as much information as possible by using symmetry arguments only.

The path integral formalism of QFT appears to be the most suitable one. We introduce^(20,21,22) the generating functional

$$\begin{aligned} W[J, j, n] = & \frac{1}{N} \int [d\psi][d\psi^+] \exp \left\{ i \int d^4x \right. \\ & \times [\mathcal{L}[\psi(x)] + J^+(x) \psi(x) + \psi^+(x) J(x) \\ & \left. + j^+(x) D_{\psi}^{(-)}(x) + D_{\psi}^+(x) j(x) + D_{\psi}^{(3)}(x) n(x) - i \in D_{\psi}^3(x) \right\} \end{aligned} \quad (3.43)$$

$D_{\psi}^{(i)}(x)$ ($i = +, -, 3$) is the dipole density operator, whose explicit form is not needed here, but whose transformation law under $SU(2)$ group is

$$D_{\psi}^{(i)} \rightarrow D_{\psi}^{(i)}(x) - \lambda_j \varepsilon_{ijk} D_{\psi}^{(k)}(x) \quad (i, j, k = 1, 2, 3) \quad (3.44)$$

Notice that $D_{\psi}^{(\pm)}(x) = D_{\psi}^{(1)}(x) \pm iD_{\psi}^{(2)}(x)$ The number N in Eq. (3.43) is

$$N = \int [d\psi][d\psi^+] \exp\{i d^4x [\mathcal{L}[\psi(x)] - i\varepsilon D_{\psi}^{(3)}(x)]\} \quad (3.45)$$

The fields ψ and ψ^+ and the sources J and J^+ anticommute, while j and j^+ commute. The ε term in (3.43) takes care, as usual, of the vacuum non-vanishing polarization:

$$\langle 0 | D_{\psi}^{(3)}(x) | 0 \rangle = P \neq 0 \quad (3.46)$$

The parameter ε has to be put to zero only *after* all the mathematical manipulations have been performed. Equation (3.46) is referred to as the spontaneous breakdown of the $SU(2)$ dipole rotational symmetry.

The ground state does not share the Lagrangian full invariance, but it is invariant only under rotations around the third axis (i.e., the direction of the electret). This group of residual invariance transformations is a subgroup of the original $SU(2)$ symmetry group. The nonvanishing electric polarization (3.46) is the trigger of the Fröhlich regime.

In Refs. 20, 21, 22 a detailed mathematical treatment is given which illustrates the consequences of the condition (3.46) on the formalism derived from the generating functional (3.43). Here we sketch only the main points and the conclusions. The first point is the validity, in the absence of long-range fields (gauge fields), of the Goldstone theorem. Two massless boson modes $P(x)$ and $P^+(x)$ must appear under the condition (3.46). The set of “phenomenological” fields [namely, the solution of Eq. (3.2)] in our case is then the field $\phi(x)$ of the free excitation quanta and the Goldstone fields $P(x)$ and $P^+(x)$:

$$\Lambda(\partial) \phi(x) = 0 \quad (3.47)$$

$$\left[i \frac{\partial}{\partial t} + \omega(\partial) \right] P(x) = 0 \quad (3.48)$$

The form of the differential operators $\Lambda(\partial)$ and $\omega(\partial)$ depends on the details of the dynamics and cannot be given here. Of course, more excitation fields can be added, but here we are interested *in* the Goldstone modes only. In particular we leave for further research the effects of the finite temperature, which imply the consideration of thermal excitations.

Let us concentrate on the Goldstone modes. They are the necessary

consequence, through the Goldstone theorem, of the nonvanishing vacuum polarization (water electret). The interaction of these modes with the “matter” field ϕ is embodied in the scattering matrix S :

$$S(\phi, \phi^+, P, P^+) = \langle : \exp[-iA(\phi, \phi^+, P, P^+)] : \rangle \quad (3.49)$$

while the dynamical map (3.3) is expressed in a path integral formalism as

$$SD^{(i)}(x, \phi, \phi^+, P, P^+) = \langle D_\psi^{(i)}(x) : \exp[-iA(\phi, \phi^+, P, P^+)] : \rangle \quad (3.50)$$

and

$$\begin{aligned} A(\phi, \phi^+, P, P^+) = & \int d^4x [\rho^{-1/2} P(x) K(\vec{\partial}) D_\psi^{(-)}(x) \\ & + \rho^{-1/2} D_\psi^{(+)}(x) K(-\vec{\partial}) P^+(x) \\ & + z^{-1/2} \phi^+(x) A(\vec{\partial}) \psi(x) \\ & + z^{-1/2} \psi^+(x) A(-\vec{\partial}) \phi(x)] \end{aligned} \quad (3.51)$$

ρ and z are the wave function renormalization constants, $\vec{\partial}$ and $\vec{\partial}$ denote space-time derivatives operating on the right and left side, respectively, and

$$K(\partial) = - \left[i \frac{\partial}{\partial t} + \omega(\partial) \right] \quad (3.52)$$

By standard techniques it is possible to get the rules of transformation of the fields ϕ and $P(P^+)$ when the field $\psi(x)$ undergoes an $SU(2)$ transformation (3.42):

(a) case $\theta_2 = \theta_3 = 0$:

$$\begin{aligned} \phi(x) &\rightarrow \phi'(x) = \phi(x) \\ P(x) &\rightarrow P'(x) = P(x) + i\theta_1 \left(\frac{P}{2} \right)^{1/2} \\ P^+(x) &\rightarrow P'^+(x) = P^+(x) - i\theta_1 \left(\frac{P}{2} \right)^{1/2} \end{aligned} \quad (3.53)$$

(b) case $\theta_1 = \theta_3 = 0$:

$$\begin{aligned} \phi(x) &\rightarrow \phi'(x) = \phi(x) \\ P(x) &\rightarrow P'(x) = P(x) - \theta_2 \left(\frac{P}{2} \right)^{1/2} \\ P^+(x) &\rightarrow P'^+(x) = P^+(x) - \theta_2 \left(\frac{P}{2} \right)^{1/2} \end{aligned} \quad (3.54)$$

(c) case $\theta_1 = \theta_2 = 0$:

$$\begin{aligned}\phi(x) &\rightarrow \phi'(x) = e^{(1/2)i\theta_3\sigma_3}\phi(x) \\ P(x) &\rightarrow P'(x) = e^{-(1/2)i\theta_3}P(x) \\ P^+(x) &\rightarrow P'^+(x) = e^{+(1/2)i\theta_3}P^+(x)\end{aligned}\tag{3.55}$$

The generators of the above three transformations are, respectively, in the space of physical states,

$$\begin{aligned}D^{(1)} &= \left(\frac{P}{2}\right)^{1/2} \int d^3x [P(x)f(x) + P^+(x)f^*(x)] \\ D^{(2)} &= -i \left(\frac{P}{2}\right)^{1/2} \int d^3x [P(x)f(x) - P^+(x)f^*(x)] \\ D^{(3)} &= \int d^3x \left[\phi^+(x) \frac{\sigma_3}{2} \phi(x) - P^+(x)P(x) \right]\end{aligned}\tag{3.56}$$

The function $f(x)$ is a square-integrable c -number solution of the free field equation (3.48):

$$K(\partial)f(x) = 0\tag{3.57}$$

and makes the integrals (3.56) convergent. The condition (3.57) makes also the generators (3.56) time independent, so that the corresponding transformations can be physically implemented without any expense of energy.

In other words the transformations (3.53)–(3.55) generated by the operators (3.56) are invariant transformations of the free-field “phenomenological” equations (3.47) and (3.48) for any choice of $f(x)$. Notice that the $P(x)$ field “translations” in Eqs. (3.53) and (3.54) are understood as the limit for $f(x) \rightarrow 1$ of the transformation generated by $D^{(1)}$ and $D^{(2)}$, namely,

$$P(x) \rightarrow P'(x) = \lim_{f(x) \rightarrow 1} \left[P(x) + i\theta_1 f(x) \left(\frac{P}{2}\right)^{1/2} \right]\tag{3.58}$$

The transformations (3.53)–(3.55) do not belong to the original $SU(2)$ group of symmetry [see Eq. (3.42)] but to the so-called $E(2)$ group—the group of the transformations of the Euclidean two-dimensional space into itself. The $E(2)$ group is the group “contraction” of $SU(2)$. We have shown explicitly that the group of symmetry at the level of observable quantities is

different from the group of symmetry at the level of the basic fields. This difference vanishes when the vacuum net polarization P vanishes.

Let us show now that the vacuum electric polarization is induced by a coherent homogeneous condensation of the massless dipole-wave quanta $P(x)$ in the vacuum $|0\rangle$, which is controlled in turn by the field “translation” (3.53) and (3.54). We get indeed, by using Eq. (3.56),

$$\langle 0 | e^{-iD^{(2)}} e^{-iD^{(1)}} P^+(x) P(x) e^{iD^{(1)}} e^{iD^{(2)}} | 0 \rangle = P |f(x)|^2 \xrightarrow{f \rightarrow 1} P \quad (3.59)$$

We define

$$|f\rangle = e^{iD^{(1)}} e^{iD^{(2)}} |0\rangle \quad (3.60)$$

Then Eq. (3.59) becomes

$$\langle f | P^+(x) P(x) | f \rangle = P |f(x)|^2 \xrightarrow{f \rightarrow 1} P \quad (3.61)$$

By introducing the annihilation operator $P_{\bar{K}}$ of a quantum of the field $P(x)$ with momentum $\bar{K}$ and taking the Fourier amplitude $f_{\bar{K}}$ of $f(x)$, we get

$$P_{\bar{K}} |f\rangle = P^{1/2} f_{\bar{K}} |f\rangle \quad (3.62)$$

which shows that $|f\rangle$ is a coherent state for the $P(x)$ field. In conclusion, a coherent homogeneous (i.e., not space dependent) condensation of dipole-wave quanta $P(x)$ is responsible for the ordered state of polarization. No expense of energy is required since $D^{(1)}$ and $D^{(2)}$ are time independent. We can then identify the dipole-wave Goldstone fields $P(x)$ and $P^+(x)$ with the coherent electric polarization waves predicted by Fröhlich.

A relevant feature of this theory is the so-called “low-energy theorem,” analogous to corresponding theorems which hold in ferromagnetism and in particle physics. The invariance of the “phenomenological” field equations under the transformations (3.53)–(3.55) requires a corresponding invariance of the S matrix. In the case of an infinite volume [namely, $f(x) \equiv 1$], this requirement is shown to imply

$$S(\phi(x), \phi^+(x), P(x), P^+(x)) = S(\phi(x), \phi^+(x), \partial P(x), \partial P^+(x)) \quad (3.63)$$

The S matrix, then, would be independent of $P(x)$ in the zero-momentum limit; in the infinite volume case, the Fröhlich regime would be stable against external perturbations exciting soft (zero momentum) modes.

In realistic systems, however, the volume is finite and surface effects become important. For a finite volume V it has been shown⁽⁴⁾ that

$$\begin{aligned}
 \langle 0 | D_{\Psi}^{(3)}(x) | 0 \rangle &= \lim_{\varepsilon \rightarrow 0} i\varepsilon \int_{-1/\eta}^{1/\eta} d^3x \int \frac{d^3p}{2\pi} e^{i\mathbf{p} \cdot \mathbf{x}} \rho(p) \\
 &\quad \times \left(\frac{1}{\omega(p) - i\varepsilon a} - \frac{1}{\omega(p) + i\varepsilon a} \right) \\
 &= \lim_{\varepsilon \rightarrow 0} i\varepsilon \int d^3p \delta_{\eta}(p) \rho(p) \left(\frac{1}{\omega(p) - i\varepsilon a} - \frac{1}{\omega(p) + i\varepsilon a} \right) \\
 &= \lim_{\varepsilon \rightarrow 0} i\varepsilon \rho(\eta) \left(\frac{1}{\omega(p=\eta) + i\varepsilon a} - \frac{1}{\omega(p=\eta) - i\varepsilon a} \right) \quad \text{at } \eta \sim 0
 \end{aligned} \tag{3.64}$$

where $\eta = V^{-1/3}$, $\delta_{\eta}(p)$ approaches $\delta(p)$ as $\eta \rightarrow 0$ and the following relation has been used:

$$\lim_{\eta \rightarrow 0} \int d^3p \delta_{\eta}(p) f(p) = f(0) = \lim_{\eta \rightarrow 0} \int d^3p \delta(p - \eta) f(p)$$

The function $\omega(p)$ is the energy of a dipole-wave quantum of momentum p ; for $V \rightarrow \infty$ ($\eta \rightarrow 0$) $\omega(p=0)=0$. Then, as $V \rightarrow \infty$, or, alternatively “far from the boundaries” of the system, $\omega(p=\eta) \rightarrow 0$ and Eq. (3.64) reduce to Eq. (3.46) [notice that $\rho(0)=p/2$]. Near the boundaries (η different from zero), $\omega(p=\eta)$ behaves as an “effective mass” of the quantum of the $P(x)$ field.

Consequently the range of the correlation mediated by $P(x)$ and the domain of the coherence will extend over a distance η^{-1} . Finite (but long) range correlation forces are then expected in realistic (finite volume) systems. A similar conclusion can be drawn from Eq. (3.58) when an appropriate cut-off function $f_{\eta}(x)$ is introduced. In conclusion a relation between the size of the system and the not exactly zero mass of the P -field quantum is obtained. As announced in Section 2 a more correct treatment, which considers the electric field as a gauge field, would lead to a dynamically constructed (nonzero) mass of the quanta, producing thus a finite volume for the biological system.

Let us now look at the form the low-energy theorem assumes for $\eta \neq 0$. The field equation becomes

$$K_{\eta}(\partial) P(x) = 0 \tag{3.65}$$

The operator $K_\eta(\partial)$ depends upon the effective mass. Then the transformation rule of $P(x)$ changes to

$$P(x) \rightarrow P(x) + f_\eta(x) \text{ const} \quad (3.66)$$

where $f_\eta(x)$ is prescribed to be a c -number solution of Eq. (3.65). In Eq. (3.63) $\partial P(x)$ must be replaced by $K_\eta(\partial) P(x)$ in order to have an S matrix invariant under the $E(2)$ group transformations. Thus an energy threshold, equal to the effective mass of the quantum, appears, through $K_\eta(\partial)$, in the S matrix. The consequence is that external perturbations whose energy is below threshold cannot perturb the system. Since the effective mass increases as $1/\eta$, the smaller the system the higher the threshold. The existence of an energy threshold matches a corresponding feature of the Fröhlich theory.

4. Phenomenological Evidence

In the above dynamical scheme a crucial role is played by solitons, water electrets, and coherent electric waves. In this section we present some experimental findings which provide evidence for the existence of such collective modes in biological systems.

4.1. Solitons on Macromolecules

Support for solitons on different kinds of molecules has been accumulated in these last years; α -helix proteins have been extensively investigated.⁽³⁾ It has been recognized that the hydrogen-bond filaments along α -helix chains support a pathway for solitons in the nonlinear dynamics of the chains. It has been shown that a simple positive anharmonicity in the interaction among nearest-neighbour monomers is the required nonlinearity in the chain dynamics (C=O bonds in α -helix).^(27,28)

A measure of the dipole-dipole interaction energy between two adjacent monomers and of the anharmonicity of the chain can be obtained by an inspection of the bandwidth of the absorption band of the monomer excitation, that forms the exciton. For α -helix proteins in which inhomogeneous broadening has been eliminated by ordering processes of the samples, the bandwidth of the C=O absorption at 1660 cm^{-1} ⁽²⁹⁾ gives evidence of the excitonlike collectivization of the vibrational C=O excitation along the chain. This is a prerequisite for the existence of Davydov solitons on the chain.⁽²⁾

In recent years, it has been suggested that Davydov solitons could actually occur on a number of one-dimensional chains of biological

interest. Raman experiments on green algae *Chlorella P.*⁽³⁰⁾ suggested the possible formation of Davydov solitons on carotenoid chains.⁽³¹⁾ Infrared absorption studies of acetanilide, a model system for one-dimensional chains of H bonds, have been recently interpreted in terms of Davydov-like solitons, involving hydrogen-bonded amide groups.^(28,32) Of particular interest is the possible existence of solitons on nucleic acids.^(19,33–35) Whether Davydov solitons could actually occur on DNA, has been examined.⁽¹⁹⁾ The stretching vibration of the P–O bond in the phosphodiester backbone of the double helix, being directed along the axis of the chain, satisfies the symmetry and polarity requirements for a collectivization of the excitation along the chain. However, only DNA in B-form displays the proper anharmonicity for the formation of a vibrational P–O exciton. This can be seen from the width of the P–O Raman band at 830 cm^{-1} ^(36,37) For DNA in A-form the band of the P–O stretching is shifted at 807 cm^{-1} and narrowed in a sharper line, showing that the P–O excitation remains localized on individual P–O groups and does not spread along the chain in exciton form. A prerequisite for the appearance of Davydov solitons on DNA is then fulfilled only for B helix structures. Different biological functions are associated to B (resting) and A (active) DNA; a soliton dynamics could have a major role in DNA structural and functional transitions. Indeed, $B \rightarrow A$,⁽³⁵⁾ $B \rightarrow Z$ ⁽³⁴⁾ transitions as well as the opening of DNA helices⁽³³⁾ have been described in terms of soliton dynamics.

4.2. Water in Biological Systems

Water in biological systems exhibits peculiar properties. Only a percentage of total water appears in the form of usual liquid water.^(38,39) A relevant part is made up of water molecules which are bound to different sites of macromolecules in the form of hydration water, while a sizable amount, although not fixed to any definite molecular site, is strongly affected by macromolecular fields; this kind of water has been termed “vicinal” water.⁽³⁹⁾ “Vicinal” water has been extensively investigated in the last decade and it exhibits noticeable properties.⁽⁴⁰⁾ It does not have a unique freezing temperature but an interval ranging from -70 to -50°C ; it is a very bad solvent for electrolytes, but nonelectrolytes have the same solubility properties in it as in usual water. Its viscosity is enhanced and its NMR response is anomalous. The whole of these physical properties point to the existence of some dynamic correlation among its constituents.

It is interesting to notice that most of the “vicinal” water surrounds the elements of the cell cytoskeleton.⁽³⁹⁾ It is not unlikely that the coherent electric waves discussed in this paper are responsible for such a correlation.⁽²⁰⁾ Bound water also exhibits some kind of collective behavior;

actually it has been shown that bound water produces electrets around many macromolecules. Activation energies of about 0.3–0.4 eV have been detected and relaxation times up to 10^5 s have been measured, suggesting the existence of zero-frequency modes of oscillation.^(9,10) A zero-frequency mode, i.e., in our theoretical framework a massless mode, appears when dynamical rearrangements of symmetry are at work. Since the electret activation energy is of the same order of magnitude as the energy required for a Davydov soliton, we think that the electret is the result, via a spontaneous symmetry breakdown, of the soliton regime occurring on biomolecules.

It is worthwhile noticing that hysteresis has been observed in all cases^(9,10); in our model then the electret would appear different in subsequent cycles (alternance of Davydov and Fröhlich regimes) during the cell lifetime, giving rise to a nonstationary evolution instead of a periodic one (on a time scale comparable with the lifetime of the biological system under consideration).

4.3. Coherent Electric Waves in Living Matter

Up to now, different kinds of evidence can be quoted, which support the Fröhlich proposal of electric waves in living matter. First of all, we mention those resonance experiments where external electromagnetic waves of selected frequencies interact with biological targets inducing macroscopic nonthermal consequences. There is a growing literature^(41–43) about microwave interaction with different types of cells. It has been observed that at given frequencies the effects occur only when a threshold is overcome and are independent from different intensities of irradiation.

A similar phenomenon, as far as the biological effects are concerned, has been observed in the visible region for a long time,⁽⁴⁴⁾ but only recently has satisfactory phenomenological evidence been achieved.⁽⁴⁵⁾ In this case, astonishingly, there is only one frequency in the visible domain (at 632.8 nm, the wavelength of He–Ne lasers) which is the same for different types of cells (for instance *E. coli* and He–La). Moreover, active cells emit electric and electromagnetic radiation on a very large frequency spectrum. Measurements have been performed at frequencies lower than 100 GHz^(46,47) and in the optical range.⁽⁴⁸⁾ This last type of emission, which could be of coherent type, has been attributed to the dynamics of DNA in the cell.

Further evidence for coherent electric waves comes from Raman experiments on living cells.⁽⁴⁹⁾ These experiments show a time-dependent pattern of lines in IR and IR regions. The connections between Raman spectroscopic data and the underlying (spatial and temporal) order in living matter is discussed elsewhere.^(50–52)

A different kind of support for Fröhlich ideas has been brought by experiments on rouleaux formation in erythrocyte suspensions.⁽⁵³⁾ These experiments show that selective long-range interactions appear among metabolically active red blood cells. The connection between the experimental outcome and Fröhlich theory has been recently discussed.^(20,54)

In conclusion, phenomenological evidence supports the main elements of our QFT approach. We feel that the task of getting a satisfactory dynamical description of living systems is not out of reach.

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